CarVision: offline car recognition in Tashkent

CSAI-3210 Introduction to Artificial Intelligence | Webster University Tashkent
Midterm report | Fall 2026 | CarVision 2.12 TEST | 28 October 2026

1. Team, repository and contributions

Student: Sofia Rakhmankulova. The user reports that Sofia performed all student project work; no second contributor has been identified. Reported responsibilities: project coordination, collecting local photographs, running the application and reviewing errors. The student must verify this attribution before submission.

Repository: https://github.com/sonix86/CarVision-2.12-Test
Release: v2.12-test. This publication is dated 28 October 2026, after the stated 27 October 23:59 deadline; no retroactive timely-submission claim is made.

2. Data: scope and provenance

The objective is a list of English (brand, colour) pairs for every visible car in a phone photograph, including Chinese EVs, multiple cars and partial views. Exact model recognition is optional. The active classifier has 82 brand labels and a separate 43-family head. The 325-name catalogue is a requested scope, not 325 trained models; several sparse brands cannot obtain a confirmed prediction under calibration.

The audited brand feature corpus contains 3,851 car crops: 2,205 training, 704 validation and 942 development-test records. It combines public Stanford Cars and attributed Wikimedia/Flickr vehicle photos with a small local phone subset. Public data dominate, so the requirement to rely primarily on own Tashkent phone data has not yet been satisfied.

The current reviewed local export contains 44 objects in 43 available photographs, 13 brands and three outing/source groups; all are TRAIN-only. Missing original files are excluded. This export overlaps historical training data and must not be added to 3,851 as an independent new sample count. Six newer reviewed training objects supplied 36 augmented feature views for the scoped correction head.

Labels use source metadata, visible badges, original-image review and user corrections. An early GAC Aion attribution was corrected to Dongfeng/Aeolus E70 after comparison with independently labelled source images. Predictions are never accepted as ground truth. Some imported public labels remain provisional; the audit does not certify every visual label.

Reviewed local training examples



BYD | grey



Zeekr | grey



Leapmotor | white

Examples shown above were used for development/training; they are not a blind test. Original provenance remains in data/reviewed_owner_v212/labels.jsonl.

2. Data distribution and split (continued)

All 82 labels in checkpoint order. T = training; V = validation; D = reused development-test. Counts are cached crop records, not unique physical vehicles. Brand indices also identify the confusion matrix on page 5.

#	Brand	T	V	D	#	Brand	T	V	D
1	Aeolus	3	0	0	42	Lada	16	3	3
2	Aito	14	7	6	43	Leapmotor	27	10	6
3	Arcfox	9	4	3	44	Li Auto	35	11	10
4	Audi	78	26	35	45	Lotus	12	4	5
5	Avatr	18	8	5	46	Lucid	9	4	2
6	BAIC	2	0	0	47	Luxeed	10	5	4
7	BMW	51	14	47	48	Lynk & Co	12	4	5
8	BYD	149	39	38	49	M-Hero	10	3	2
9	Baojun	10	4	5	50	MG	16	8	4
10	Bestune	4	1	1	51	Maextro	5	3	1
11	Buick	17	7	6	52	Mazda	4	2	3
12	Cadillac	11	5	4	53	Mercedes-Benz	70	25	36
13	Changan	13	6	5	54	Neta	19	12	9
14	Chery	57	17	12	55	Nio	25	12	8
15	Chevrolet	255	51	50	56	Nissan	42	17	13
16	Ciimo	8	4	7	57	Onvo	10	5	3
17	Daewoo	64	13	13	58	Ora	13	4	4
18	Deepal	20	6	10	59	Pocco	2	0	0
19	Denza	13	5	3	60	Polestar	11	4	4
20	Dongfeng	23	8	9	61	Porsche	10	4	4
21	Everus	8	3	5	62	Rivian	8	2	3
22	Exlantix	14	6	5	63	Roewe	13	5	4
23	FangChengBao	7	2	2	64	Rolls-Royce	8	4	4
24	Ford	60	20	30	65	Skywell	4	3	1
25	Forthing	11	6	4	66	Smart	10	5	3
26	GAC Aion	27	10	8	67	Stelato	10	5	4
27	GMC	9	4	4	68	Tank	5	2	1
28	Geely	31	11	8	69	Tesla	54	24	16
29	Geometry	23	5	5	70	Toyota	98	28	94
30	Hanteng	3	0	2	71	Volkswagen	72	25	34
31	Haval	44	11	6	72	Volvo	60	20	30
32	HiPhi	9	3	4	73	Voyah	12	4	5
33	Honda	46	14	43	74	Weltmeister	18	4	5
34	Hongqi	14	6	11	75	Wey	4	3	1
35	Hyptec	15	4	6	76	Wuling	17	4	4
36	Hyundai	140	28	153	77	XPeng	34	11	12
37	IM Motors	15	7	4	78	Xiaomi	10	6	4
38	JAC	5	5	6	79	YangWang	2	1	1
39	Jetour	3	1	0	80	Zeekr	56	20	13
40	Kia	30	7	7	81	Zotye	6	4	2
41	LEVC	4	2	1	82	iCAR	9	4	2

Total: 3,851 = 2,205 T + 704 V + 942 D. Source-prefix counts: make_scope_v23 1407, public_prototype 625, stanford 634, images 528, ev_v19 194, ev_patch_v22 202, ev_patch_flickr_v22 26, make_flickr_v23 9, own_oct21 17, ev_views_v26 156, owner_oct25 40, e70_truth_v30 13.

Public subsets retain recorded groups; matching source hashes and exact pixels/masks must remain in one split. The 2.12 structural audit found no group, source-hash or exact-pixel cross-split leakage in these 3,851 records. Near-duplicates and repeated physical vehicles without an identity may remain. Repeated use for model development prevents calling D an independent final test.

3. Architecture

Multi-car instance masks isolate body evidence from streets and neighbouring cars. General visual backbones supply complementary shape/detail descriptors; small supervised heads are feasible to adapt on CPU. This is the actual 2.12 route, rather than the legacy MobileNet-only design.

Phone-to-answer pipeline
Phone photo -> local resize / optional tiles and mirrored view
COCO Mask R-CNN ResNet50-FPN-v2 -> visible car/truck masks
Mask-aware crop -> frozen DINOv2 ViT-S/14 + frozen CLIP ViT-B/32
768-D DINO descriptor + 512-D CLIP -> normalized 1,280-D features
Own 82-brand cosine head + grouped reference fallback + calibrated gates
Own 43-family head; scoped make correction for Zeekr / Leapmotor / BYD
Parallel own colour ensemble + masked paint / illumination evidence
Per-instance English (brand, colour) list -> GUI / JSON / TXT / annotated JPG

Each cosine head learns class vectors, biases and a logit scale. Probability, margin, absolute similarity, crop geometry and grouped validation support determine whether a result is confirmed. The family head operates with the accepted make; its manufacturer fallback is limited to Dongfeng when the primary make abstains. Low confidence yields unknown or a visibly unconfirmed hypothesis, not a guaranteed answer.

Parameter counts: the make head has 105,043 trainable parameters; the 43-family head has 55084; the additional scoped make head has 105043. DINO encoder state contains 22,056,576 elements. The paired CLIP visual state contains 87849216 elements (state counts, not the whole application). Encoders and detector are frozen in the deployed classifier training.

Colour is independent of make: own VCoR-trained CNN/statistical heads predict eight classes, then visible-paint and scene-illumination evidence are fused. White, black, grey, red, blue, green, brown and yellow are the output vocabulary. Warm light, reflections and windows still cause errors.

Originality and changes in 2.12

Only general-purpose COCO/DINO/CLIP initialization is used. No third-party ready-made car-brand or car-colour classifier and no external inference API are used. Architectural uniqueness relative to classmates must be checked by the student/instructor.

Version 2.12 fixes the detector-only route selecting the wrong classifier, default colour loading, the joint-colour API, paired-encoder hash checks, conflicting labels/duplicate splits and unsafe reuse of a training output directory. Recognition weights are unchanged from 2.11. These engineering fixes are not evidence of an accuracy increase.

4. Training and 5. Evaluation

Training used PyTorch CPU execution with three worker threads in the logged head-fitting scripts. Frozen features were cached. Weighted cross-entropy counters imbalance; label smoothing and public-teacher KL distillation limit forgetting. Views include original/mask-aware crops, synthetic phone degradation, horizontal flips, darker colour casts and family edge cuts. Augmentations stay with their source group.

Stage	Recorded configuration
82-brand adaptation	AdamW; LR 0.0001 joint; wd 0.02; CE smoothing 0.01; local weight 12; public KL 0.2; selected epoch 80 / 100
43-family adaptation	Only three corrected family vectors adapted; AdamW; round 1 LR 0.0001; selected epoch 100; other vectors frozen
Scoped correction	Two 80-epoch fits, batch 128; LR 0.00005 / 0.0001; wd 0.01; smoothing 0.01; local weights 12 / 16; KL distillation
Masked paint head	AdamW 0.001; wd 0.03; batch 64; 160 epochs; smoothing 0.03; selected epoch 116; recorded 532.7 seconds

Validation chooses candidates and calibrates rejection. Both unrestricted new make replacements failed guards; only the calibrated scoped correction was retained. The original encoders were not retrained. The aggregate elapsed time for the complete development history was not recorded; the paint-head time above is one component, not total training time.

Brand: complete routed classifier, crop-level results

Cohort / view	N	Raw top-1	Correct / N	Wrong	Unknown
Validation / whole	704	81.0%	432/704	6	266
Validation / synthetic phone	704	72.4%	331/704	6	367
Development-test / whole	942	83.1%	626/942 (66.5%)	23	293
Development-test / synthetic phone	942	76.5%	478/942 (50.7%)	24	440
Target-make subset / whole	174	76.4%	90/174 (51.7%)	4	80
Target-make subset / synthetic phone	174	63.8%	47/174 (27.0%)	6	121

Correct/N counts abstentions as failures. Raw top-1 ignores rejection. Confirmed precision on D/whole is 626/649 = 96.5%, which is not 96.5% overall recognition. The 20-make target subset includes mixed-powertrain manufacturers and is not 174 certified pure EVs. All cohorts are reused development evidence; synthetic phone views are not new real phone photographs.

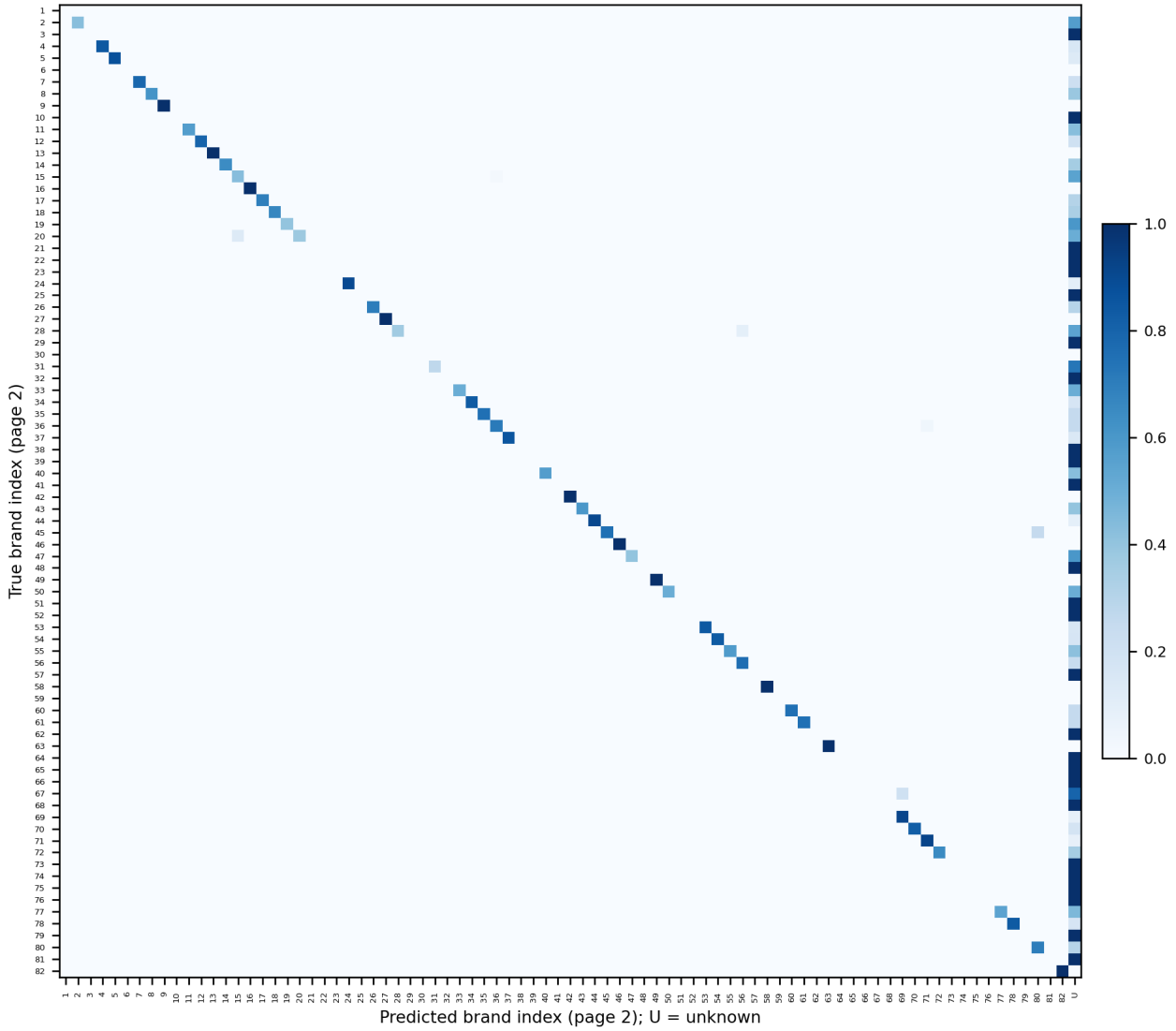
Colour data and component validation

Colour	white	black	grey	red	blue	green	brown	yellow
VCoR train	399	404	785	633	738	558	1585	1108
VCoR validation	84	86	167	135	154	120	337	236
VCoR test	82	84	164	133	156	119	339	237
Reviewed local TRAIN	20	11	12	0	1	0	0	0

Recorded source-domain colour ensemble: validation 94.24% (1,319 crops), test 93.61% (1,314). Masked-paint component: validation 93.75% (160), test 92.08% (240). These are component results, not current end-to-end Tashkent colour accuracy. No independent current full-pipeline colour validation has been completed.

5. Confusion, phone tests and error analysis

Current routed make confusion on all 704 validation crops, whole view; row-normalized. All 82 classes are retained, including sparse classes; zero-support rows are blank. Numeric class labels map to page 2; U is abstention. Full integer counts are supplied in the companion CSV. This matrix was regenerated from the active 2.12 checkpoint.



Chinese-brand failures are driven by similar sedan/SUV bodies, limited local rear/night views, small or absent badges, crop/background sensitivity and sparse group support. The E70/GAC mix-up also exposed a reference-label mistake. A high similarity bar is not a calibrated probability of correctness; forcing the top class would convert unknowns into confident mistakes.

Known real-photo diagnostics: Zeekr and Leapmotor make corrections now pass, but the exact Zeekr 007 model is not trained. Voyah and a white Chevrolet Tracker remain wrong/unconfirmed; AIQAR is outside the 82-brand vocabulary. A white BYD Dolphin may be coloured brown. Some parked cars remain too small or occluded. These known cases became training/development evidence and cannot establish generalization.

Historical local rehearsal: 99 photographs were processed, but only a 41-object reviewed subset had usable make/colour references. Version 2.10 achieved 37/41 confirmed makes and 26/41 colours on that known development subset. This is not a 2.12 blind-test score or accuracy over all 99 photos. No defensible prediction of “8 out of 10 new photos” is available.

Engineering verification: 150 unit tests passed; final EXE analysis of six known photographs took 14.8-16.3 seconds per process on this PC, excluding transfer. EXE help, crop selection/export, paired weights and the detector-only route were checked with Python removed from PATH. A second physical PC and a full timed exam rehearsal have not yet been tested.

6. How to run and 7. Reflection

Exact test-day procedure

1. Extract the complete CarVision 2.12 Full archive into a new writable Windows 10/11 x64 folder. Keep CarVision.exe, models and _internal together. Do not install old patches over it. No Python installation is needed for the packaged application.
2. Run CHECK_ON_NEW_PC.cmd once, then disable Wi-Fi and verify a photograph. PASS checks execution, not answer accuracy.
3. Launch CarVision.exe and confirm the title says 2.12 TEST. Transfer the exam photo by the permitted local method; USB avoids dependence on Telegram/internet.
4. Choose Open photo, then Analyse photo once. Review every visible-car row. Copy the English brand/colour pairs using the assignment-answer button; do not substitute a speculative alternative as a confirmed result.
5. Record the answer before seeing the reference. Preserve JSON/TXT/annotated output. Measure the complete transfer-to-answer cycle against the exam time limit.

Command-line alternative inside the application folder

```
CarVision.exe predict "C:\photos\car.jpg" --mode fast --output outputs\exam
CHECK_ON_NEW_PC.cmd
```

For development only, after installing requirements: python launch.py predict "C:\photos\car.jpg" --mode fast --output outputs\exam. The older EXAM/midterm wrapper is optional and is not the primary GUI path. Personal training must write to a fresh output folder; it does not automatically update the advanced 82-brand head. Training scripts and recorded selection policies document the specialist lineage.

Reflection and AI-use statement

With two more weeks, the priority is independently verified local phone data: distinct vehicles, makes, front/rear/side views, occlusion and evening light. Reserve an untouched vehicle/session-separated test before further fitting. Train balanced body and badge/detail specialists with hard negative pairs, improve visible-paint/illumination handling, and report detection recall, make/colour accuracy, unknown rate and the actual exam score. These are proposed improvements, not implemented results or an 80% guarantee.

OpenAI Codex assisted with requirements analysis, code implementation, debugging, experiment scripts, packaging, automated verification and preparation of this report. Human-provided photos, feedback and corrections were used, but some labels required independent rechecking. Sofia must review and understand the submitted code, verify the contribution statement and report actual work honestly. No AI assistant or external service is called during inference.

Publication: https://github.com/sonix86/CarVision-2.12-Test. Code, release restore instructions, trained weights, available data/crop caches, licences and evidence are supplied. The midterm tag identifies the real publication snapshot, not a claim of submission before the deadline. Original local history is preserved without invented or backdated commits. Student understanding and architectural uniqueness still require instructor verification.

References and reproducibility evidence

[1] Assignment: CSAI-3210 Midterm, Fall 2026, sections 4-7; locally extracted in outputs/assignment-source.txt.

[2] Generic detector: pytorch.org/vision/stable/models/mask_rcnn.html; DINOv2: github.com/facebookresearch/dinov2; CLIP visual initialization: huggingface.co/laion/CLIP-ViT-B-32-laion2B-s34B-b79K.

[3] Stanford Cars: ai.stanford.edu/~jkrause/cars/car_dataset.html. Wikimedia/Flickr author, licence and source records: dataset manifests and THIRD_PARTY.md. VCoR: kaggle.com/datasets/landrykezebou/vcor-vehicle-color-recognition-dataset. Observe each source licence; do not publish private photos without permission.

[4] Measurements: models/phone_v32/regression.json; outputs/ev-specialist-v212/baseline.json; models/colour_training.json; models/masked_colour_training.json. Current validation matrix regenerated by tools/report_v212.py.

[5] Integrity: outputs/pipeline-audit-v212/audit.json; outputs/tests-audit-v212-final.log; outputs/exe-v212-verification/verification.json. Active make SHA256: c1b01c33d55e163815734ba8c37810da97ac02bba0f9d042519635815d9fac14.